The following report is an advance copy of the case narrative on the Khamisiyah Ammunition Storage Point. This advance copy contains the text of the narrative with endnotes. This initial Gulflink posting will be expanded within the next few weeks to include copies of the supporting documentation, hyperlinked to the endnotes. The present version, however, is intended to make the text of the narrative immediately available to the general public.

U.S. DEMOLITION OPERATIONS AT THE KHAMISIYAH AMMUNITION STORAGE POINT

Case Narratives are reports of what we know today about specific events that took place during the Gulf War of 1990 and 1991. This particular case narrative focuses on the actions of American troops at Khamisiyah. In addition, we report on when it became known that American troops may have been exposed to chemical agents there. This is an interim report, not a final report. We hope that you will read this and contact us with any information that would help us better understand the events reported here. With your help, we will be able to report more accurately on the events surrounding Khamisiyah. Please contact my office to report any new information by calling:

1-800-472-6719

Bernard Rostker
Special Assistant for Gulf War Illnesses
Department of Defense

Last Update: February 21, 1997

Many veterans of the Gulf War have been experiencing a variety of physical symptoms, collectively called Gulf War illnesses. In response to veterans' concerns, the Department of Defense (DoD) established a task force in June 1995 to investigate all possible causes. On 12 November 1996, responsibility for these investigations was assumed by the Investigation and Analysis Directorate (IAD), Office of the Special Assistant for Gulf War Illnesses (OSAGWI) which has continued to investigate the events that occurred at Khamisiyah. Its interim report is contained here. In addition, the Army Inspector General was directed by the Secretary of the Army on 25 September 1996 to conduct an investigation into Army operations at Khamisiyah, and the Assistant to the Secretary of Defense for Intelligence Oversight was directed by the Deputy Secretary of Defense on 25 September 1996 to review what the intelligence communities knew concerning Khamisiyah. These independent efforts have not yet been completed and may shed additional light on events at Khamisiyah.

As part of the effort to inform the public about the progress of this effort, DoD is publishing on the Internet and elsewhere accounts related to possible causes of Gulf War illnesses, along with whatever documentary evidence or personal testimony was used in compiling the account. The narrative that follows is the first such account.

SUMMARY

The story of the Khamisiyah Ammunition Storage Point or ASP has three parts: the efforts of U.S. forces to destroy Khamisiyah, the inspection of the site by the United Nations Special Commission or UNSCOM, and the public inquiry into the events that occurred there, "what we knew, and when we knew it:"

The Destruction of Khamisiyah

Immediately following the end of Operation Desert Storm, U.S. Army units occupied the area known as Objective GOLD and later identified as the Khamisiyah ASP (which was also known as Tall al Lahm or Suq Ash Shuyukh). Khamisiyah was a huge ammunition storage site, covering 50 square kilometers and containing about 100 ammunition bunkers and several other types of storage facilities. The XVIII Corps (Airborne) (ABN) dispatched combat engineer and demolition units to Khamisiyah to destroy its munitions and facilities.

To perform the demolition, U.S. forces set off two very large explosions, one on 4 March 1991 and a second on 10 March 1991. They also set off a number of smaller explosions to destroy small caches of munitions and to test techniques for destroying bunkers. Demolition operations continued in the Khamisiyah area through most of April 1991.

During the demolition operations, and, indeed, throughout the entire period of U.S. occupation at Khamisiyah, there were no reports of verified chemical agent detections, nor were there reports of anyone, soldier or civilian, experiencing symptoms consistent with exposure to a chemical agent.

Inspecting Khamisiyah

In October 1991 and March 1992, and then again in May 1996, the UNSCOM inspected Khamisiyah, specifically searching for chemical weapons. Based on their own inspections and information provided by the Iraqis, UNSCOM inspectors identified three sites in and around Khamisiyah that had contained chemical weapons: in an area that became known as the "pit;" in Bunker 73, one of the bunkers subsequently identified as having been blown up by U.S. troops; and in an above-ground storage area.

In October 1991, UNSCOM inspectors found about 300 damaged and intact 122mm rockets in an area surrounded by a berm southeast of the main ASP. This area became known as the "pit." Their investigation showed that the intact rockets contained chemical agents (sarin and cyclosarin). During a subsequent visit in March 1992, about 500 rockets were blown up on site near the "pit", with the remaining rockets being shipped to Al Muthanna, Iraq for subsequent destruction. The UNSCOM destruction efforts accounts for 782 rockets; the Iraqis report that 2,160 such rockets had been at Khamisiyah. It is unknown how many of the unaccounted for rockets were destroyed by U.S. forces.

During the 1991 inspection, the Iraqis claimed that chemical munitions found in the "pit" had been salvaged from Bunker 73 and that both had been destroyed by Coalition Forces. UNSCOM inspectors visited the site of the bunker, which appeared damaged, and used chemical agent monitors. These monitors were negative, and the inspectors did not thoroughly search the bunker.

The UNSCOM team was also shown an above-ground storage site about 3 kilometers west of the ASP containing 6,300 intact 155mm artillery shells filled with mustard agent. To date, there is no evidence that any Coalition Forces had been to this site. These rounds were also shipped to the destruction facility at Al Muthanna.

US intelligence became aware of the UNSCOM findings in November 1991, but at the time this report did not result in identification of which, if any, U.S. troops participated in demolition activities at Khamisiyah. The lack of contemporaneous U.S. reports of chemical weapons, and the fact that the Iraqis were selective in their willingness to cooperate, as reported by UNSCOM to the United Nations Security Council, led to the belief the Iraqis were not telling the truth about chemical weapons being at the site when the demolition occurred. In May 1996, UNSCOM again returned to Khamisiyah, where the team conclusively identified debris in the rubble of Bunker 73 that was characteristic of chemical munitions.

The Public Inquiry

In February 1994, a request from Congressman Browder to the UN for any reports about chemical weapons found in Iraq after the Gulf War rekindled U.S. interest in Khamisiyah. The UN responded with a letter in April 1994 which listed Khamisiyah along with other chemical weapons sites. During hearings on export administration in May 1994 before the Senate Banking, Housing and Urban Affairs Committee, DoD witnesses admitted the UN had found chemical weapons at Khamisiyah but were unable to confirm that any U.S. troops were at the site.

In March 1995, as a result of Presidential concerns, the CIA began a reexamination of relevant intelligence. In May 1995, a Presidential Advisory Committee (PAC) was created. In June 1995, DoD formed the Persian Gulf Illnesses Investigation Team (PGIIT). Throughout 1995 and 1996, interest in Khamisiyah and the events surrounding it increased. On June 21, 1996, DoD confirmed publicly that "U.S. soldiers from the 37th Engineer Battalion destroyed ammunition bunkers at [Khamisiyah] in early March 1991 ... it now appears that one of these destroyed bunkers contained chemical weapons."

DoD investigation into the subject continues. What follows provides additional detail about the events described in this summary. The information upon which this narrative is based is incomplete. As the investigation continues, the IAD hopes to answer a number of these questions, including the following:

- How many chemical warfare munitions were present at Bunker 73 and at the "pit" at the time the U.S. demolitions took place?
- Were two separate groups working in the "pit" on 10 March 1991?
- Was there an additional demolition of munitions at the "pit" on 12 March 1991?
- Who were the 15 to 20 engineers assigned to assist the EOD noncommissioned officer in the "pit" on 10 or 12 March 1991?
- What were the weather conditions on the day(s) of the "pit" demolition(s)?

NARRATIVE (An acronym listing/ glossary is at Tab A)

Introduction

The Khamisiyah ASP, also known to Coalition Forces as Tall al Lahm, Suq Ash Shuyukh (local Iraqi place names), or Objective GOLD [1] , was a large munitions storage depot. It is located in southern Iraq along the southern side of the Euphrates River and about 25 kilometers southeast of the city of An Nasiriyah. The ASP area borders a major highway [2] used extensively by U.S. troops transiting the area after the cease-fire began. Khamisiyah was an extensive complex of above- and below-ground ammunition bunkers, general storage buildings, and open equipment storage revetments (sand mounds, or berms) covering approximately 50 square kilometers. The main site covered 25 square kilometers. Figure 1 shows the location of Khamisiyah in the Kuwait Theater of Operations (KTO).

Beginning in late 1995, both the U.S. Intelligence Community and DoD's Persian Gulf Illness Investigation Team (PGIIT) began a thorough review of Iraqi chemical capabilities during Operations Desert Storm/Desert Shield and the demolition of munitions at the Khamisiyah ASP. These investigations eventually led DoD to announce that "it now appears that one of these destroyed bunkers contained chemical weapons. [3] ." The following details what is currently known of the events at Khamisiyah ASP involving U.S. troops:

Desert Storm Activities

At the opening of the Gulf War (January 1991), the U.S. Central Command (USCENTCOM) did not classify Khamisiyah as a chemical weapons storage site [4] . However, by late February 1991, the XVIII Corps (ABN) G-3 indicated that Khamisiyah was suspected of being a chemical weapons storage site [5].

During the Air War of Operation Desert Storm (16 January - 1 March 1991), Coalition Force aircraft attacked Khamisiyah [6], destroying scores of warehouses and several ammunition bunkers [7]. At the commencement of the Ground War (24 February 1991), it was widely believed that U.S. Forces operating in the KTO after G-Day were likely to capture chemical warfare (CW) and, possibly, biological warfare (BW) munitions of various types [8]. Accordingly, all command levels issued Commander's Guidance for Disposition of Captured Chemical and Biological Munitions and other directives for dealing with captured Iraqi CW or BW munitions (see USCINCCENT on 24 February 1991 [9], COMUSARCENT on 21 February 1991 [10] , and XVIII Corps (ABN) on 27 February 1991 [11]). For example, the Commander, 24th Infantry Division (Mechanized) (ID(MECH) [12]), also issued a memorandum on 16 February 1991 detailing the guidance for handling these items. The handling/disposition of CW or BW munitions guidance documents emphasized safety and security for both Coalition Forces and the local population:

> Destruction of munitions or bulk agent will be accomplished in accordance with established EOD field disposal policies and procedures to ensure the complete and safe destruction of the captured items. Prior to destruction, all necessary measures to preclude collateral damage or down-wind hazard to friendly forces and civilians will be accomplished [13].

Destruction of Munitions at Khamisiyah ASP

The XVIII Corps (ABN) had the mission to conduct movement to contact operations, including attacking and securing Objective GOLD (later identified as Khamisiyah). On 26 February 1991, the first US troops to reach Khamisiyah were from the 24th ID(MECH).

> On the northern end of BP 102, LTC John Craddock maneuvered his 4-64th Armor Battalion toward a canal north of Highway 8....Continuing north, the battalion overran a huge, untouched ammunition storage area and pushed the beaten Iraqis protecting the facility into the weeds near the canal [14].

On 26 February 1991, the 24th ID(MECH) received information from the XVIII Corps (ABN) that there were "possible chemicals on Objective GOLD [15]." On 27 February 1991, the 24th ID(MECH) secured Objective GOLD [16] and continued eastward beyond Khamisiyah to cut-off retreating Republican Guard divisions near Basrah. On 28 February 1991, the 82nd Div (ABN) was located west of the 24th ID (MECH) with the "3rd Brigade conduct[ing] movement to Objective GOLD [17];" the Objective was secured on 1 March 1991 [18]. Although there is no evidence to date that the 82nd Div (ABN) received the warning from the XVIII Corps (ABN) of possible chemicals on Objective GOLD, in reporting activities that occurred in securing Khamisiyah, the 82nd Div (ABN) Chemical Officer noted that standard procedures were followed:

> When the 82nd Div (ABN) initially occupied the sector, FOX vehicles and unit reconnaissance teams checked for evidence of contamination or chemical weapons. No contamination was found. Riot control agent CS was found in the Tall al Lahm ASP.....White phosphorus [artillery] rounds were also found. Artillery rounds with fill plugs and central bursters were found. They were marked with a yellow band. They were empty. Other rounds in the area were marked similarly. FOX reconnaissance determined they [the rounds] contained TNT [19].

On 1 March 1991, the 2nd Platoon, Charlie Company, 307th Engineer Battalion, in direct support of TF 2-505, part of the 82nd Div (ABN), reconnoitered Khamisiyah ASP and concluded that demolition operations would require additional engineer support. Subsequently, the 37th Engineer Battalion was told to destroy the approximately 100 bunkers at Khamisiyah ASP [20].

On 2 March 1991, the XVIII Corps (ABN) noted:

> XVIII ABN Corps continues defensive/ security operations in zone with emphasis on force protection, clearing of residual enemy personnel in sector and destruction/evacuation of captured enemy equipment. Now that the tempo has dropped, units are able to begin clearing bunker complexes that were initially bypassed to maintain momentum. Divisions are discovering large numbers of bunkers/underground complexes containing weapons, ammunition and other materials. Destruction of these bunkers has already begun; however, the enormity of the task before us and amount of resources required is still unknown.
>
> ***
>
> Commander's evaluation... Our emphasis is on protection of the force and operations [21].

Early on 2 March 1991, a platoon from Charlie Company, 37th Engineer Battalion arrived at the Khamisiyah ASP as an advance party for the battalion [22]. Upon its arrival, the unit found a large number of the local civilians and many animals inside the ASP; many were inside the bunkers as well [23].

On 3 March 1991, the remainder of the 37th Engineer Battalion (-) [24]and two teams (three soldiers each) from the 60th Explosive Ordnance Disposal Detachment (EOD) arrived at Khamisiyah [25]. The battalion had M8A1 chemical alarms mounted on various unit vehicles, and these were reported to be operational [26]. The battalion's chemical noncommissioned officer (NCO) stated he was in "MOPP 4" [27]and checked [28]some of the bunkers for chemical agents. The results of these checks were reported to be negative. As part of the operation, the U.S. troops searched the site for any "special" weapons, that is chemical weapons and laser- or optically-guided munitions. They found one rocket with possible intelligence value; all remaining were deemed conventional [29]. Two bunkers (98 and 99) were exploded to test demolition techniques [30].

On 4 March 1991, the three line companies of the 37th Engineer Battalion, assisted by the two teams of the 60th EOD, were each assigned 12 to 14 bunkers to inventory and demolish [31]. According to the Charlie Company Commander, "the explosive ordnance guys came through and said, here's what you're looking at. These are safe to destroy." [32] Therefore, the engineers planned to use the explosives necessary to destroy conventional munitions. A total of 38 [33]bunkers were rigged with explosives, including the bunker subsequently reported by the Iraqis as containing chemical munitions (Bunker 73). Reports and interviews [34]indicate that approximately 300 engineer and EOD personnel participated in the demolition at the ASP, and about 770 additional personnel from the 505th Infantry secured the area.

At approximately 1400 hours on 4 March 1991, 37 of the 38 bunkers exploded (explosives in Bunker 92 failed to go off due to a bad time fuse) [35]. The weather was clear, with winds coming from the SW [36]. The engineer battalion set up an observation point approximately 3 to 4 kilometers northwest, and crosswind of the Khamisiyah ASP (see unit location on Figure 2).

At approximately 1445 hours on 4 March 1991, an M8A1 chemical alarm in Bravo Company, 37th Engineer Battalion sounded at the observation point. Since troops were at MOPP 0 [37], upon hearing the alarm, some went to MOPP 4 status, and others only donned their masks [38]. Each company and EOD team [39]performed several M256 kit tests [40]. Two NBC NCOs interviewed say they got "weak" or "slightly" positive results on M256 tests, although the test kit is designed to show either positive or negative results. The Bravo Company Commander observed the test performed by his NBC NCO and states he saw a negative result, not a "weak positive." The second NBC NCO states he did a second test that was negative [41] [42]. An "all clear" was then signaled. Interviews of medical personnel at battalion/brigade/division/corps-level did not reveal any evidence of symptoms or health

problems related to chemical warfare agent exposure during the entire period in question [43]. Debris from the exploding bunkers (described as fragments, and in some instances intact weapons) landed in or near the observation point, so troops were moved further away from Khamisiyah [44].

On 5 March 1991, there were heavy rains in the morning, and many vehicles became stuck. The 60th EOD teams examined the bunkers from the previous day's demolition and determined one bunker (92) did not explode. The explosives were re-fused and set off without incident [45]. EOD reviewed the results of the previous day's demolitions and decided to use a different technique to destroy the remaining bunkers [46]. Alpha Company of the 307th Engineer Battalion [47] was given the mission to destroy warehouses in the NW portion of Khamisiyah ASP. The XVIII Corps (ABN) SITREP for this day also noted that the 82nd Div (ABN) destroyed ASPs at Jalibah and Tallil. There is no mention of Khamisiyah or Objective GOLD [48].

On 6 March 1991, each engineer company of the 37th Engineer Battalion and Alpha Company from the 307th Engineer Battalion exploded a bunker to test the latest techniques for demolition developed by the 60th EOD. The EOD experts attempted to implode the bunkers [49] in order to reduce the number of secondary explosions and to conserve the amount of explosives used. During 7-9 March 1991, no demolitions were performed because of poor weather. The time was used for demolition training, rehearsals, and inventorying [50]the remaining bunkers and warehouses.

On 9 March 1991, the Operations Officer of the 37th Engineer Battalion found crates of 122mm rockets outside the SE corner of Khamisiyah ASP [51]. A noncommissioned officer from the Headquarters & Headquarters Company (HHC) of the battalion was told to destroy these munitions in what is now called the "pit" area of Khamisiyah [52].

On 10 March 1991, at approximately 1540 hours, crates of rockets in the "pit" were detonated. At the same time, the 60 remaining bunkers were detonated by 37th Engineer Battalion, and the warehouses were blown up by Alpha Company of the 307th [53]. There is some confusion as to whether the HHC NCO with a two-man detail was the only group setting explosives in the "pit." Photo analysis of the "pit" reveals 13 separate stacks of material. The HHC NCO and one of his detail both state they rigged 3 stacks of rockets for demolition, no other stacks were observed, and no one else was working in the "pit." However, an EOD NCO says he led a 15 - 20 man engineer/soldier detail that destroyed approximately 850 rockets (6 to 8 stacks) in the "pit" on the same day as the "big explosion" on March 10 [54].

An accounting of demolition at the "pit" is also noted in the 60th EOD log [55]for 12 March 1991. It was recorded that 840 "5-inch" (this measure approximates 122mm) rockets were destroyed at coordinates for Khamisiyah ASP. This report, however, conflicts with information provided by an NCO from the 60th EOD [56] [57].

The 37th Engineer Battalion observation point for the demolition on 10 March 1991 was south of Khamisiyah on MSR 8, approximately 20-30 minutes travel time by vehicle away from the ASP. Once they heard explosions, the 37th continued south towards Saudi Arabia [58] for approximately four more hours. The weather was overcast skies with poor visibility; wind direction and speed on this date are the subject of ongoing investigation by the Institute for Defense Analyses (IDA) and CIA.

On 12 March 1991, the 307th Engineer Battalion [59]identified additional ammunition stores southwest of Khamisiyah ASP, described as "another enemy bunker complex of more than 400 revetted bunkers with large caches inside." [60]During the period 15-19 March 1991, the 307th Engineer Battalion rigged explosives on the munitions found in the berm area southwest of Khamisiyah ASP. On March 20, the berm area was detonated at approximately 1530 hours [61].

On 23 March 1991, the 2nd Armored Cavalry Regiment, part of the U.S. VII Corps, assumed responsibility for the area of operations, which included Khamisiyah. The 84th Engineer Company and the 146th EOD were among their supporting units. On 24 March 1991, the 82nd Div (ABN) [62], the 307th Engineer Battalion, and the 60th EOD departed for Saudi Arabia and subsequent redeployment [63].

On 27 March 1991, the 2nd Armored Cavalry Regiment was told to determine if Tall al Lahm Ammo Storage Depot South (100 revetments) and Tall al Lahm Ammo Storage Facility [Khamisiyah] contained possible chemical/biological munitions. On 28 March 1991, the unit reported to VII Corps that chemical/biological reconnaissance of both Tall al Lahm sites yielded negative results [65].>

On 2 April 1991, the 82nd Engineer Battalion, located south of the area of operations, reported hearing a large explosion in the vicinity of Tallil, another site of demolition, approximately 40 km from Khamisiyah [66].

On 6 April 1991, members of the 84th Engineer Company and 146th EOD re-examined bunkers at Khamisiyah ASP, and determined that six bunkers required additional detonations to destroy remaining munitions [67].

The last American units departed Khamisiyah in late April 1991.

Further details on this chronology are being gathered in the continuing investigation by the Investigation and Analysis Directorate of the Office of the Special Assistant for Gulf War Illnesses.

UNSCOM Investigations at Khamisiyah

In April 1991, the UN Security Council adopted Resolution 687, setting specific terms for a formal cease-fire to end the conflict between Iraq, Kuwait and the countries cooperating with Kuwait [68]. In May 1991, in response to UN Security Council Resolution 687, the Iraqis declared to UNSCOM that "Khamisiyah (Nasiriyah)" was a chemical weapons storage site, although it was not included in their first declaration to the UN in April 1991. This was confusing information because it referred to two locations, a known site (Nasiriyah), and an as yet unknown site (Khamisiyah).

In October 1991, UNSCOM sent a team to inspect six of the sites which were not near Baghdad. The site map provided to the UNSCOM Team was labeled "An Nasiriyah Depot S.W. (Khamisiyah)," and it depicted the layout of what U.S. Intelligence knew as An Nasiriyah ASP. However, the UNSCOM Team was not taken to An Nasiriyah, but to a different site, which is now known to be Khamisiyah. They were shown artillery shells and rockets in two separate areas apart from the main ASP (see Figure 2). An open area, 3 kilometers west of the bunkers, contained 6,323 155mm artillery shells filled with mustard agent. These shells were undamaged and were stored in an orderly fashion (in several stacks/clusters) under tarpaulins, using the natural terrain features to hide them. The second area, located in a "pit" south of the main bunker complex, contained 297 122mm rockets in three to four "heaps," some of which were damaged but most were intact. Some rockets were neatly laid out, while others appeared to have been bulldozed into piles or heaps. Many rockets were leaking, and plastic inserts and other features characteristic of chemical munitions were observed, so UNSCOM personnel drilled into one of the intact rockets to take a sample. The sample was later analyzed and found to be a chemical warfare nerve agent (sarin/cyclosarin (GB/GF)).

The Iraqis told UNSCOM in 1991 that chemical rockets found in the "pit" had been salvaged from Bunker 73, which had been destroyed as part of the demolition operations by Coalition Forces. UNSCOM acknowledged that Bunker 73 appeared damaged, but did not thoroughly inspect the bunker. Chemical agent monitoring at the bunker site was negative. No other observations were documented concerning remains of munitions, such as whether there were observable plastic inserts or other paraphernalia characteristic of chemical munitions.

In November 1991, the U.S. Intelligence Community became aware of the results of the UNSCOM Khamisiyah Ammunition Storage Facility site visit [69]. The U.S. Intelligence Community did not believe Iraqi accounts to the UN that chemical weapons had been blown up at Khamisiyah by the coalition forces at the end of the war [70]. They believed the Iraqis were engaged in possible deception, consistent with the observations of UNSCOM in their inspections and analysis of Iraqi declarations [71].

Despite their doubts, intelligence analysts initiated a search for any U.S. units involved in blowing up munitions at Khamisiyah. A response to their request dated 12 November 1991 indicates that they had "received information from ARCENT [the Army Central Command] to the fact that 24th Mechanized Infantry Division was located in the vicinity of Tall al Lahm, but [were] unable to confirm if U.S. troops did in fact destroy buildings at this particular site." [72] ARCENT mistakenly identified the 24th Infantry Division as being in the area at the time, although they had not carried out the demolition at Khamisiyah. The ARCENT lead was followed, and a 20 November 1991 message notes that "Info on Tall al Lahm Ammo Depot was passed to ... G-2 Office, Ft. Stewart, GA," Headquarters of the 24th Mechanized Infantry Division. Further, this message states "info on presence of troops there and their activities during Desert Storm were requested...." [73] The IAD has followed that lead; after more than five years, the person contacted at Fort Stewart has no specific recollection of being contacted or of any specific subsequent actions taken. Additional follow-up has provided no further leads at this point [74].

During a March 1992 visit, the UNSCOM Team consolidated and destroyed at least 500 122mm rockets. According to the UNSCOM press release [75].

on 30 March 1992, the munitions destroyed included full, partially full, and empty rockets. This number includes the 297 rockets mentioned previously, which were found in the "pit". In addition to the rockets destroyed in the March 1992 site visit, more than 200 [76].

rockets were unearthed by the Iraqis in the "pit" and shipped to Al Muthanna for destruction. More than 700 rockets or major rocket parts in all were found in the "pit" area. The actual number of rockets in the "pit" and Bunker 73 is unknown, and continues to be topic of questioning during interviews with 1-800 callers and other interviewees.

The Public Inquiry

In February 1994, Congressman Browder (D-AL) requested from the UN any reports pertaining to chemical weapons found in Iraq after the Gulf War. The UN responded by letter in April 1994, providing in tabular format a listing of the sites at which Iraqi chemical warfare agents/weapons were found. Included in this listing was the "Khamisiyah Storage Site." [77].

In May 1994, witnesses from DoD testified before the Senate Banking, Housing, and Urban Affairs Committee (the Riegle Committee) on matters relating to export administration. In the course of that testimony, DoD witnesses acknowledged that the UN had found chemical munitions at a site, 15 nautical miles from An Nasiriyah, but stated that U.S. forces were not at that site, which they said was north of the Euphrates River [78]. Review of the testimony and responses to questions for the record submitted by DoD in September and October 1994 reveals that there was true confusion as to the location of Khamisiyah and its proximity to US troops. Furthermore, DoD believed that any destruction of chemical munitions at this "other site" (Khamisiyah) probably had occurred after the war as part of an Iraqi deception campaign. [79].

This belief formed the basis for information provided to the Defense Science Board Task Force Persian Gulf War Health Effects in June 1994. The Task Force report stated that:

> There were also reports of damage by the United Nations Special Commission inspection team that visited a different location in the general vicinity of An Nasiriyah several months after the cessation of hostilities. There are indications that the site visited by the UNSCOM team was not a site targeted during the air war but may have been specially constructed for the UN inspectors.

In November 1994, Congress directed the expansion of a DoD Gulf War registry, to include all service members [81]. The agency tasked with responsibility for compiling the unit locator database was the Environmental Support Group (ESG) (now referred to as the U.S. Armed Services Center for Research of Unit Records). The ESG unit locator database incorporates all available coordinates (both latitude/longitude and universal trans-mercator indices) derived from unit logs, situation reports, etc. It reports the location of many, but not all, of the U.S. units in Iraq and Saudi Arabia during the conduct of the Gulf War by unit identification codes (UICs) and time.

In March 1995, the President directed [82] a more intensive effort to discover the causes of illnesses among Gulf War veterans. As concern over the Gulf War illnesses mounted the Acting Director Central Intelligence directed the CIA to conduct a comprehensive review of relevant intelligence information. In this review the CIA focused on identifying and quantifying Iraqi chemical, biological, or radiological releases during and after the war that could have reached U.S. troops [83]. As part of the President's initiative, the DoD and the CIA initiated new efforts to collect and review operational, intelligence and medical records from the war. In April, declassification of health documents started, and in June 1995, the Persian Gulf Illnesses Investigation Team (PGIIT) was established to provide a DoD organization to manage the different investigations which were now on-going.

Just prior to September of 1995, CIA analysts resurfaced the UNSCOM October 1991 Khamisiyah site visit report during a re-examination of thousands of intelligence reports and other intelligence holdings. On 6 September 1995, the CIA identified Khamisiyah as a key unresolved chemical weapons release issue, which raised special concern because its southerly location put it closest to U.S. troops. On 13 September 1995, CIA informed DoD's PGIIT of Khamisiyah's potential relevance to the exposure issue and asked whether U.S. military forces had been at the site [84]. DoD searched the newly constructed ESG unit locator database and indicated that some units were in the area. In October 1995, PGIIT learned from the ESG that the 37th Engineer Battalion reported a location coordinate near Khamisiyah, but there was no indication of their mission. At that time, no follow-on investigation into the 37th Engineer Battalion activities was conducted.

The CIA continued to monitor the DoD's Khamisiyah investigation and to conduct their own research. On 26 January 1996, the CIA briefed the National Security Council (NSC) staff that U.S. troops probably blew up chemical weapons at Khamisiyah. The Presidential Advisory Committee (PAC) [85],

formed in May 1995, was subsequently made aware of these initial findings. DoD and the CIA began an intense and comprehensive effort to research and analyze the Khamisiyah events. Concern about U.S. exposure increased as the topic became more fully understood. By early March 1996, CIA and PGIIT pieced together previously unanalyzed information indicating activity at the Khamisiyah ASP, and, for the first time, they received clear indications that the 37th Engineer Battalion blew up Bunker 73 at Khamisiyah.

On 10 March 1996, a CIA analyst heard a tape recording of a radio show during which a veteran (Mr. Brian Martin) of the 37th Engineer Battalion described demolition activities at a facility the analyst immediately recognized as Khamisiyah [86]. Although Mr. Martin had previously testified before the House Veterans Affairs Committee and had been contacted by DoD after the release of the Riegle report [87], it was not until the CIA analyst heard the 10 March 1996 broadcast that the possible connection between An Nasiriyah demolitions and the bunkers at Khamisiyah was drawn. DoD and the PAC were notified of this connection on 11 March 1996.

A PGIIT investigator contacted Mr. Martin on 11 March 1996 about the demolition he had witnessed, and, with assistance from the PAC, Mr. Martin provided a video tape that showed the demolition activities he had witnessed. Another version of the tape [88] confirms the event on Mr. Martin's tape as the demolition at Khamisiyah on 4 March 1991. Review of these tapes has provided much useful information to the investigation by confirming events and weather data. Unfortunately, no such video, photographs or logs have been found that document the 10 March 1991 demolition.

On 1 May 1996, the CIA publicly announced at a PAC hearing that UNSCOM had found chemical weapons at Khamisiyah and, that "elements of the 37th Engineer Battalion.... performed demolition of munitions at this facility" during 1991 [89].

On 14 May 1996, UNSCOM again visited Khamisiyah. During this visit, the Iraqis told the inspectors that the 6,323 mustard rounds had been moved to Khamisiyah from Al Muthanna to An Nasiriyah in January 1991 after the beginning of the Gulf War. The Iraqis further stated that about 2,160 sarin/cyclosarin rockets were also brought from Al Muthanna in January 1991, and stored in Bunker 73 until a chemical leak was discovered, causing approximately 1100 of the rockets to be moved to the "pit" area in February 1991. According to the Iraqis, this was done before the Coalition Forces destroyed the ammunition storage area.

On 21 June 1996, DoD held a news briefing to detail these findings on Khamisiyah. The DoD said:

> UNSCOM has informed us that, as part of its ongoing effort to verify Iraqi declarations, it inspected the Khamisiyah ammunition storage area last month [May 1996]. During that inspection, UNSCOM concluded that one bunker had contained rockets with chemical agents. U.S. soldiers from the 37th Engineer Battalion destroyed ammunition bunkers at this site in early March 1991, shortly after the war ended. Based on a new review of the available information, it now appears that one of these destroyed bunkers contained chemical weapons [90].

After the 21 June 1996 announcement, the focus of investigation shifted to better understand two questions. First, what was the potential for exposure to chemical agents at Khamisiyah, and second, who might have been exposed. DoD merged the ESG unit locator database with DMDC personnel databases to identify the people actually deployed at varying distances from Khamisiyah ASP in early March 1991 [91]. Efforts are on-going to identify additional units and individuals which were in the vicinity of Khamisiyah (see Tab B to this document). In addition, the PGIIT, CIA, and DMDC conducted interviews with U.S. troops known to be involved in the demolition to try to reconstruct such information as the exact dates of the demolition, amount and type of munitions destroyed, and weather and wind direction on the dates of demolition.

Potential for Exposure - Plume Analysis

The CIA was charged by the PAC [92] to develop predictions of the potential chemical fallout from the March 1991 demolition operations using, among other models, the U.S. Army's Chemical and Biological Defense Command's NUSSE4 transport and diffusion model. The results were briefed to the PAC on 9 July 1996, and on 2 August 1996, the CIA published a report on the Bunker 73 explosion on 4 March 1991. They concluded that the likely movement of vapor was to the east and northeast away from U.S. troops [93].

With regard to the "pit " explosion, the CIA encountered numerous modelling uncertainties, especially weather data, and could not come to any definitive conclusions. On 29 October 1996, DoD asked IDA to convene an independent panel of experts in meteorology, physics, chemistry, and related disciplines to review all of the modelling efforts available in order to determine the potential fallout from the "pit" area demolition. IDA provided a progress report on 18 December 1996. At that time, IDA reported:

> .... continued concern about the inability to describe the many variables of the agent-munition release mechanism. The panel agrees with the CIA that "huge uncertainties remain" in the number of rockets present for destruction and the number of those rockets destroyed. Among the other major variables for which there remains much uncertainty are total quantity of agent released, mechanism of release, and purity of agent [94].

The expert panel is working with DoD investigators and was briefed by CIA analysts in order to assess the model inputs and to determine whether the original dispersion and weather models (or any other models) may be effective in predicting the possible extent of chemical exposure as a result of the Khamisiyah demolitions.

Who Was At Khamisiyah

On 7 August 1996, the Assistant Secretary of Defense for Health Affairs designed and conducted a telephone outreach to veterans who may have participated in the operation at Khamisiyah ASP. Based on a search of the ESG database and over 100 interviews, the PGIIT was able to determine units potentially involved in this operation. Individuals were selected for the telephone outreach based on their Gulf War assignment to one of these units.

DMDC identified 1179 individuals assigned to units thought to have participated in the operation. Of those identified, 542 individuals were contacted and completed the survey, 14 were uncooperative with telephone operators, and 12 individuals are deceased. The telephone outreach effort concluded in October 1996. All individuals who were not able to be contacted via the telephone were mailed a certified letter, informing them of the incident and

requesting they share any information pertaining to the incident through the 1-800 hotline. 259 individuals received the certified letter but did not contact DMDC, and 352 individuals have yet to receive a letter because either it is in the process of being forwarded to them or they have no known address.

The personal descriptions of the incident offered by each individual completing the survey were analyzed to screen for potential leads for the continuing investigation. The PGIIT used the data as a basis for follow-up interviews. Of the total 542 contacted, 39 individuals mentioned chemical alarms sounding during this period. These 39 reports, and all subsequent reports of chemical alarms sounding, are the subject of continuing examination and further analysis by the investigators of the IAD, the successor organization to PGIIT.

Given the uncertainty concerning the fallout from the "pit" demolition on 10 March 1991 and after careful review of the CIA's preliminary results, DoD decided to be conservative and notify all those who were thought to be within a 50 kilometer radius of Khamisiyah ASP between 1 March and 15 March 1991. Letters were sent to approximately 21,000 Gulf War veterans. The intent of these letters was to inform them of the incident; to inform them of the potential for low-level exposure to chemical warfare agent; to explain how to sign up for examination in the DoD or Department of Veterans Affairs registries; and to notify them of a forthcoming survey to query for specific unit/individual location information, chemical exposure data, and health and medical program participation questions. The most important part of the letter was:

> We need to hear from you, not only about your experience in the vicinity of the site, but also about any health problems you think may be a result of your service during Operations Desert Storm/Desert Shield. Your timely response to the survey will provide us with critical information. If you have information that you believe would be of immediate value to us pertaining to the events at Khamisiyah, please call the PERSIAN GULF INCIDENT HOTLINE at 1-800-472-6719.
>
> If you are experiencing health problems you believe to be a result of your service in Operation Desert Storm/Desert Shield and you are eligible for health benefits through the Department of Defense, please call the COMPREHENSIVE CLINICAL EVALUATION PROGRAM at 1-800-796-9699. If you are eligible for benefits provided by the Department of Veterans Affairs system, please call the PERSIAN GULF HELPLINE at 1-800-PGW-VETS [95].

Mailing of the survey started 10 January 1997 and is still continuing [96].

This case is still being investigated. As additional information becomes available, it will be incorporated. If you have records, photographs, recollections, or find errors in the details reported, please contact the DoD Persian Gulf Task Force Hot Line at 1-800-472-6719

END NOTES

[1] Objective GOLD was a military designation for the area around what was then referred to as the Tall al Lahm ASP. GOLD was an Objective for the 24th Infantry Division (Mechanized) during the Ground War phase. The XVIII Corps Desert Shield Chronology February 1991, 26 February 1991 entry; and Brigadier General Robert H. Scales, Certain Victory, (Washington: Office of the Chief of Staff, U.S. Army, 1993), Figure 5-1.

[2] This highway was referred to as "Highway 8" or "MSR [military supply route] 8." It became the major redeployment route to reach MSR "Texas" and "Virginia," which then led back into Saudi Arabia and the units' assembly areas. 20th EN Bde General Update and Unit Location Report, 3 March 1991.

[3] DoD News Briefing, 21 June 1996.

[4] DoD News Briefing, 21 June 1996. Since Khamisiyah was not specifically listed as a suspected chemical weapons storage site, it was considered to be a conventional weapons storage site. CIA Timeline on Activities Involving Khamisiyah Depot, for June 1996 PAC briefing.

[5] XVIII Corps CTOC 26 February 1991 log entry, and supporting handwritten action message form.

[6] Khamisiyah was targeted 10 times, however, only 8 missions were completed; 5 were B-52G raids and 3 were attacks by fighter/bomber aircraft. Gulf War Air Planning Staff (GWAPS) database query log.

[7] When the 82nd Airborne Division arrived at Khamisiyah on 1 March 1991, they saw evidence of this bombing destruction. Leavenworth 5+1 Press Conference video, 15 November 1996.

[8] Testimony by General (ret.) Schwarzkopf, 29 January 1997, before the Senate Veterans Affairs Committee.

[9] Commander's Guidance for Disposition of Captured Chemical and Biological Munitions, USCINCCENT, 241200Z FEB 91

[10] Iraqi Chemical Munition Disposition, COMUSARCENT, 211400Z FEB 91

[11] Captured Chemical and Biological Munitions, XVIII Corps (ABN), 270845Z FEB 91

[12] Memo, Commander, 24th ID(M), SUBJ: Destruction of Enemy Equipment and Supplies, 16 February 1991.

[13] Commander's Guidance for Disposition of Captured Chemical and Biological Munitions, USCINCCENT, 241200Z FEB 91, para. 3D.

[14] Brigadier General Robert H. Scales, Certain Victory, (Washington: Office of the Chief of Staff, U.S. Army, 1993), p. 257-259

[15] An XVIII Corps February 26,1991 log entry, and the supporting handwritten action message form.

[16] XVIII Corps (ABN) SITREP, 27 February 1991, p. 5.

[17] XVIII Corps (ABN) SITREP, 28 February 1991, p. 5.

[18] 2/505 one page history summary.

[19] 82nd Chemical Officer's handwritten message to 2nd ACR Chemical Officer, 23 March 1991, describing activities that had occurred in AO.

[20] ENSITREP, March 3, 1991. The 37th Engineer Battalion was attached to the 82nd Div (ABN) for this purpose. The 37th Engineer Battalion was tasked through its chain of command, the 937th Engineer Group and the 20th Engineer Brigade.

[21] XVIII Corps (ABN) SITREP, 2 March 1991, pp. 3-4, 8.

[22] Personal recollection of unit commander. Leavenworth 5+1 Press Conference video.

[23] Documented in interviews with soldiers present. Unit 1SG interview, Lead Sheet 843, July 1996.

[24] A (-) symbol indicates that the unit has detached part of its unit strength (personnel or units) to another area or mission. (Army manual FM 21-30, p. B-3; FM 101-5-1, Ch. 2, Sec. IV, p. 2-73). In this instance, the 37th EN Bn had begun redeploying its headquarters and much of its heavy equipment back to assembly areas in Saudi Arabia. Likewise, the 60th EOD had dispatched teams to different areas to support search and destroy operations by other 82nd DIV (ABN) units.

[25] Detailed in 37th EN Bn Operations Log, 24 February to 10 Mar 1991.

[26] Unit 1SG stated, "Each platoon had M-8 on at all times." Lead Sheet 843, July 1996.

[27] MOPP (mission oriented protective posture) ensemble is worn at certain levels, from 0 (nothing) to 4 (mask with hood, Battle Dress Overgarment (BDO), butyl rubber gloves and overshoes). (Army manual FM 17-15, App. D, Section II).

[28] These "checks" were described by the NBC NCO to consist of performing M256 kit tests.

[29] Interview with EOD NCOIC, Lead Sheet 806, June 1996.

[30] 37th EN Bn Operations Log, 24 February to 10 Mar 1991.

[31] Both the 37th EN Bn and the 307th EN Bn lacked sufficient explosives to completely destroy all the warehouses and bunkers in Khamisiyah. In order to complete the task, the engineers made use of the explosives they found on-site; most of this explosive material consisted of the Soviet version of military C-4 explosive. 37th EN Bn message, SUBJ: Time Fuze, 4 March 1991.

[32] MAJ Huber's statement on CBS Evening News, February 12, 1997.

[33] Number of bunkers rigged is based on Unit commander's personal log entries and Leavenworth 5+1 Press Conference video.

[34] Interviews with 37th EN Bn CSM and Commander, Lead Sheet 819, June 1996 and Interview Notes, June 1996.

[35] 37th EN Bn Operations Log, 24 February to 10 Mar 1991.

[36] Photograph of 4 March 1991 explosion at Khamisiyah showing flag blowing.

[37] MOPP level of protection was reduced (from level 2 to 0) based on the cessation of hostilities. The XVIII Corps (ABN) Desert Shield Chronology, February 1991, 271940Z February 1991 entry. (Higher MOPP levels were used when a unit was initially entering the bunker areas. 37th EN Bn NBC NCO interview and Lead Sheet 1094, October 1996.)

[38] EOD NCO interview, Lead Sheet 1077, October 1996.

[39] EOD NCO interview, Lead Sheet 1077, October 1996.

[40] Leavenworth 5+1 Press Conference video.

[41] Interview of NCO and commander, Lead Sheets 825 and 832, June 1996

[42] This information regarding negative detections is what is known to date and may be modified as the result of survey information. In January 1997, surveys were sent to people believed to have been within 50km of Khamisiyah, seeking additional information.

[43] Statement by Commander, 307th Medical Bn.

[44] Reports indicate fragments fell in the area for 5 to 30 minutes, and secondary explosions of munitions continued for 24 hours. Personal interviews, Unit NCO interview, Lead Sheet 1223, January 1997.

[45] Leavenworth 5+1 Press Conference video.

[46] This review of demolition techniques was, in part, prompted by a reported lack of explosives available to the engineers, concern about the amount of secondary explosions, and the extent of unexploded ordnance (UXO) caused by the March 4, 1991 demolition. Interview with unit commander, Lead Sheet 1266, January 1997, and 37th EN Bn mission update, March 1991.

[47] This unit, along with another team from the 60th EOD, arrived at Khamisiyah on 4 March 1991. Leavenworth 5+1 Press Conference video.

[48] XVIII Corps (ABN) SITREP, 5 March 1991, p.4.

[49] The test explosions did not produce the desired results. However, it was decided to change the charger method from individual bunkers to a singular ring main that included all the warehouses and bunkers. The net result would be one large explosion versus individual explosions timed to go off at

approximately the same time. Leavenworth 5+1 Press Conference video.

[50] The IAD does not have any detailed inventories of what was actually in the ASP bunkers and warehouses. Personnel on site have stated there was not sufficient time to do an exact count of munitions, and that most of the containers had Arabic writing, which was indecipherable to the troops. IAD does have an aggregate inventory report from the 307th EN Bn and 82nd DIV (ABN) reports, and videotape showing inside some bunkers. 307th EN Bn Operations Summary, and 60th EOD Incident Journal (Desert Storm)

[51] Interviews with BN S-3, Lead Sheet 1053, October 1996 and 307th Liaison Officer, Lead Sheet 1221, January 1997.

[52] HHC S-2 NCO interview, Lead Sheet 857, July 1996.

[53] Leavenworth 5+1 Press Conference video.

[54] HHC S-2 NCO interview, Lead Sheet 857, July 1996 and EOD NCO interview, Lead Sheets 910 and 1077, September and October 1996, respectively.

[55] 60th EOD Incident Journal (Desert Storm), 1 April 1991.

[56] EOD NCO interview, Lead Sheet 910, September 1996.

[57] This conflict in reports gives rise to the question of whether there was more than one "big explosion." IAD continues to seek identification of the individual soldiers involved in the demolition so as to resolve that question.

[58] The 37th EN Bn (-) continued to the assembly area (AA ELM) to link-up with the remainder of their soldiers in preparation for redeployment to Ft. Bragg. Operations Log, 37th EN Bn for 24 February to 10 March 1991.

[59] The remainder of the 307th EN Bn arrived in Khamisiyah 11-12 March 1991 from Tallil. Leavenworth 5+1 Press Conference video.

[60] XVIII Corps CTOC Journal Sheet, 12 March 1991

[61] Leavenworth 5+1 Press Conference video.

[62] Brigadier General Robert H. Scales, Certain Victory, (Washington: Office of the Chief of Staff, U.S. Army, 1993), p.326

[63] 307th EN Bn Desert Storm Narrative, 17 May 1991.

[64] VII Corps FRAGO # 189-91, 27 March 1991

[65] VII Corps Tactical Chemical Spot Report, 28 March 1991

[66] Report in unit history file states other U.S. unit was conducting demolition mission at Tallil Air Base.

[67] 84th EN Co. Commander's comments in the Leavenworth 5+1 Press Conference video.

[68] A provision of UN Security Council Resolution 687 established the UN Special Commission (UNSCOM) whose primary Objective was to identify Iraqi chemical and biological weapons and ballistic missiles which survived the war, have them moved to an Iraqi destruction facility, or to destroy the weapons themselves. UN Security Council Resolution 687

[69] Redacted Message, 12 November 1991.

[70] Mr. Denny Ross, CBS News, 12 February 1997.

[71] S/23268, Letter from Executive Chairman, Office of the Special Commission, 4 December 1991, p. 2.

[72] Redacted CIA declassified message, 12 November 1991.

[73] Redacted CIA declassified message, 20 November 1991

[74] Memorandum, XX February 1997, Discussions with the 24 ID G-2 staff.

[75] Unclassified UNSCOM Press Release,1 April 1992

[76] Reuters News Agency summary, Subject: Iraq-Chemical, 25 June 1992.

[77] Letter responding to Congressman Browder's request, UNSCOM, 5 April 1994

[78] Transcript of Hearing, Senate Banking Committee, 25 May 1994, pp. 135-137. Mr. Edwin Dorn, Under Secretary of Defense for Personnel, Dr. Theodore M. Prociv, Deputy Assistant to the Secretary of Defense for Chemical and Biological Weapons, and Dr. John Kriese, Chief Officer for Ground Forces, Defense Intelligence Agency.

[79] Responses to questions for the record submitted to Congressman Riegle on 22 September and 5 October 1994

[80] The Defense Science Board Task Force Report on Persian Gulf War Health Effects, p. 32, June 1994.

[81] Public Law 102-109, DoD to Establish PG Registry, and Public Law 102-585, Sec. 704, Expansion of Coverage of Persian Gulf Registry. The original registry was developed to identify veterans exposed to the Kuwait oil well fires.

[82] DoD News release, ref. # 116-95, 9 Mar 95.

[83] CIA Chronology of Khamisiyah Events, transmitted to Special Assistant for Gulf War Illnesses Executive Director, CIA on 24 January 1997.

[84] CIA Chronology of Khamisiyah Events, transmitted to Special Assistant for Gulf War Illnesses Executive Director, CIA on 24 January 1997.

[85] Established by Executive Order 12961.

[86] Executive Director of CIA testimony before the Senate Veterans Affairs Committee, 9 January 1997.

[87] Mr. Martin had previously testified before the House Committee of Veteran Affairs in November 1993, where he described his illness and reported that he had witnessed a scud attack, saw dead animals, took pyridostigmine, and was exposed to diesel fuel. Additionally, in May 1994, after release of the Riegle Committee report, DoD contacted him to ask if he thought he could have been exposed to chemical agents and, if so, how. He cited three possible sources of exposure: the scud attack at Wadi Al Batin, the dead animals, and smoke from a bunker destruction near An Nasiriyah.

[88] IAD obtained an original, uncut version of the videotape from Mr. Martin's Company Commander, Major Huber.

[89] Extract from Testimony of Executive Officer, Office of Weapons Technology and Proliferation, CIA, to the PAC, 1 May 1996.

[90] DoD News Briefing, 21 June 96.

[91] The ESG database is known not to be definitive. See the PAC Report, January 1997, p. 30.

[92] Statement by CIA Executive Director at News Conference on Persian Gulf Veterans Illnesses, 1 November 1996.

[93] CIA Report on Intelligence Related to Gulf War Illnesses, 2 August 1996, and DefenseLINK News Release 681-96, 20 December 1996.

[94] DefenseLINK News Release, Reference Number 681-96, 20 December 1996.

[95] Copy of letter sent to vets.

[96] Copy of survey sent to vets.

[97] Based on locations reported for battalion-level Unit Identification Codes (UICs) derived from the Geographic Information System (GIS) [UIC-based personnel strengths from the Defense Manpower Data Center(DMDC).]

TAB A - Acronym Listing/Glossary

This TAB provides a listing of acronyms found in this report. Additionally, the Glossary section provides definitions for selected technical terms which are not found in common usage.

Acronyms

1SG.....First Sergeant
AASLT.....Air Assault
ABN.....Airborne (type of unit)
ACR.....Armored Cavalry Regiment (Army unit)
ADA.....Air Defense Artillery
AMB.....Ambulance
AO.....Area of Operations
ARCENT.....Army Central Command
ASP.....Ammunition Storage Point
ATC.....Air Traffic Control
AVN.....Aviation
Bde.....Brigade (Army unit)
BDO.....Battle Dress Overgarment
Bn.....Battalion (Army unit)
CAM.....Chemical Agent Monitor
Cbt.....Combat
CCEP.....Comprehensive Clinical Evaluation Program
CENTCOM.....Central Command
CIA.....Central Intelligence Agency
Co.....Company (Army unit)
COMUSARCENT.....Commander, U.S. Army Central Command
COSCOM.....Corps Support Command
CSG.....Corps Support Group
CSM.....Command Sergeant Major
CTOC.....Corps Tactical Operations Center
DECON.....Decontamination
Det..... Detachment
DIA.....Defense Intelligence Agency
DISCOM.....Division Support Command
Div.....Division
DIVARTY.....Divisional Artillery
DMDC.....Defense Manpower Data Center
DoD.....Department of Defense (U.S.)
DVA.....Department of Veterans Affairs
EN.....Engineer (Unit designation)
ENSITREP.....Engineer Situation Report
EOD.....Explosive Ordnance Disposal
ESG.....Environmental Support Group
FA.....Field Artillery (Unit designation)

FRAGO.....Fragmentary Order
Fwd.....Forward
GA.....Georgia
GB.....Nerve agent (sarin)
GF.....Nerve agent (cyclosarin)
GWAPS.....Gulf Air War Planning Staff
HHC.....Headquarters and Headquarters Company
Hqs.....Headquarters
HTML.....Hyper-Text Media Link
IAD.....Investigation and Analysis Directorate
IN.....Infantry (Unit designation)
KTO.....Kuwaiti Theater of Operations
Maint.....Maintenance
MECH.....Mechanized
mm.....millimeter
MMC.....Materiel Management Center
MOPP.....Mission Oriented Protective Posture
MSR.....Military Supply Route
MTF.....Medical Treatment Facility
NBC.....Nuclear, Biological, and Chemical
NCO.....Non-Commissioned Officer
NCOIC.....NCO In Charge
NSA.....National Security Agency
NSC.....National Security Council
NW.....northwest
OBJ.....Objective
OSAGWI.....Office of the Special Assistant for Gulf War Illnesses
OSD.....Office of the Secretary of Defense (U.S.)
PGIIT.....Persian Gulf Illnesses Investigation Team
POL.....Petroleum, Oil and Lubricants
QM.....Quartermaster
RMC.....Regional Medical Center
S&S.....Supply and Service
SE.....southeast
SITREP.....Situation Report
Spt.....Support
Sqdn.....Squadron (Army unit)
SW.....southwest
TAC.....Tactical
TF.....Task Force
TOC.....Tactical Operations Center
U.S.....United States
UIC.....Unit Identification Code
UN.....United Nations
UNSCOM.....United Nations Special Commission
USCINCCENT.....Commander-in-Chief, U.S. Central Command
UXO.....Unexploded Ordnance

Glossary

CCEP Developed by a multi-disciplinary team of DoD and VA medical specialists, the CCEP provides a two-phase, comprehensive medical evaluation. Phase I is conducted at the local medical treatment facility (MTF) and consists of a history and medical examination comparable in scope and thoroughness to an in-patient hospital admissions evaluation. The medical review includes questions about family history, health, occupation, unique exposures in the Gulf War, and a structured review of symptoms.

Health care providers specifically inquire about the symptoms and Persian Gulf exposures listed on the CCEP Provider-Administered Patient Questionnaire. The medical examination focuses on patients' symptoms and health concerns and includes standard laboratory tests (complete blood count, urinalysis, serum chemistries) and other tests as clinically indicated.

Individuals who require additional evaluation after completing the MTF-level Phase I evaluation and appropriate consultations may be referred to one of 14 Regional Medical Centers (RMCs) for Phase II evaluations. RMCs are tertiary care medical centers that have representation from most major medical disciplines. Phase II evaluations consist of symptom-specific examinations, additional laboratory tests, and specialty consultations according to the prescribed protocol.

Reference: CCEP Report dated 2 Apr 96, can be found on homepage: http://www.tricare.osd.mil/pgulf/18k-a.html

Cyclosarin A nerve gas agent commonly referred to as GF, similar to sarin (GB) (see below), but more persistent.

References: Agency for Toxic Substances and Disease Registry (ATSDR). 1992. Toxicological profile for Cyclosarin. Atlanta, GA: U.S. Department of Health and Human Services, Public Health Service.

Detection Paper Detection paper is based on certain dyes being soluble in chemical warfare agents. Normally, two dyes and one pH indicator are used, which are mixed with cellulose fibers in a paper without special coloring (unbleached). When a drop of chemical warfare agent is absorbed by the paper, it dissolves one of the pigments. Mustard agent dissolves a red dye and nerve agent a yellow. In addition, VX causes the indicator to turn to blue which, together with the yellow, will become green/green-black.

Detection paper can thus be used to distinguish between three different types of chemical warfare agents. A disadvantage with the papers is that many other substances can also dissolve the pigments. Consequently, they should not be located in places where drops of, e.g., solvent, fat, oil or fuel can fall on them. Drops of water give no reaction.

On the basis of spot diameter and density on the detection paper, it is possible to obtain an opinion on the original size of the droplets and the degree of contamination. A droplet of 0.5 mm diameter gives a spot sized about 3 mm on the paper. A droplet/cm2 of this kind corresponds to a ground contamination of about 0.5 g/m2. The lower detection limit in favorable cases is 0.005 g/m2.

Reference: Detection of Chemical Weapons: An overview of methods for the detection of chemical warfare agents; homepage: http://www.opcw.nl/chemhaz/detect.htm

M256A1 Chemical Agent Detection Kit

The M256A1 kit is a portable, expendable item capable of detecting and identifying hazardous concentrations of chemical agent. The M256 kit is used after a chemical attack to determine if it is safe to unmask. The M256A1 kit has replaced the M256 kit. The only difference between the two kits is that the M256A1 kit will detect lower levels of nerve agent. This improvement was accomplished by using an eel enzyme for the nerve test in the M256A1 kit in place of the horse enzyme used in the M256 kit.

Reference: Worldwide Chemical Detection Equipment Handbook, p. 430

M8A1 Chemical Alarm

The M8A1 is an automatic chemical agent detection and warning system designed to detect the presence of nerve agent vapors or inhalable aerosols. The M8A1 will automatically signal the presence of the nerve agent in the air by providing troops with both a audible and visible warning. The M8A1 was fielded to replace the wet chemical M8 detector with a dry system which eliminated the M229 refill kit, the logistic burden and associated costs. The M8A1 operates in a fixed, portable, or vehicle mounted configuration.

Reference: Worldwide Chemical Detection Equipment Handbook, p. 412

Mustard

Mustard "gas" refers to several manufactured chemicals including sulfur mustard. They do not occur naturally in the environment. The term gas is in quotes because mustard "gas" does not behave as a gas under ordinary conditions. Mustard "gas" is really a liquid and is not likely to change into a gas immediately if it is released at ordinary temperatures. As a pure liquid, it is colorless and odorless, but when mixed with other chemicals, it looks brown and has a garlic-like smell. Mustard "gas" was used in chemical warfare and was made in large amounts during World Wars I and II. It was reportedly used in the Iran-Iraq war in 1984-1988. It is not presently used in the United States, except for research purposes.

The only way that mustard "gas" would enter the environment [other than through use as a weapon] would be through an accidental release. Some evaporates from water and soil into air. It does not easily go into water, and the amount that does breaks down quickly. It is more stable in soil than in water but still breaks down within days, depending on the outside temperature (cold weather makes it more stable). It does not go from soil to groundwater. Mustard "gas" does not build up in the tissues of animals because it breaks down so quickly. Mustard "gas" makes your eyes burn, your eyelids swell, and causes you to blink a lot. If you breathe mustard "gas," it can cause coughing, bronchitis, and long-term respiratory disease.

References: Agency for Toxic Substances and Disease Registry (ATSDR). 1992. Toxicological profile for mustard "gas." Atlanta, GA: U.S. Department of Health and Human Services, Public Health Service

Sarin

Sarin is a light brown liquid. It is odorless, and evaporates about as fast as gasoline. It is toxic both as fumes and to the touch. It is not as persistent an agent as Tabun or Soman, the other two of the trinity of nerve gases developed in Germany.

Sarin, along with Tabun and Soman was invented not long before the Second World War by German scientist Dr. Gerhard Schrader. While developing insecticides similar to malathion and parathion, he discovered the first "nerve gas" agents, as they were then called. In 1936 he discovered Sarin. The Germans stockpiled these weapons during the Second World War, but never used them, probably because of Hitler's personal distaste for the weapons (he himself was a victim of gas attacks in Flanders during the First World War). Sarin is now known as "GB."

Only very small amounts of Sarin are needed to kill. A single milligram of Sarin coming in contact with the skin is sufficient to kill. In a vaporous form, it takes a concentration of 100 milligrams per cubic meter to be fatal. Nerve gases such as Sarin are known as "organophosphorus anticholinesterases" or "OP's." Their chemical method of killing is to block the enzyme cholinesterase. The body's muscles receive electrical impulses caused by choline. Cholinesterase break down choline, making sure these impulses stop at the proper time. Cholinesterase attaches itself to choline and breaks it down, thus halting the impulse. Sarin fools thecholinesterase into acting upon the Sarin as it would choline. When the cholinesterase attaches itself to Sarin, it doesn't break down. Thus, choline is not broken down, and the body goes into convulsions.

The first symptoms start in the eyes, where the pupils contract and vision is blurred. It causes breathing problems and chest tightness. Finally it produces vomiting and headaches, after which the heart and lungs stop as the body convulses. The antidote is a substitute for the missing cholinesterase, which is atropine.

The armed forces in the Gulf War were given Oxime tablets in case of gas attack, which acts to release cholinesterase from the Sarin.

References: Agency for Toxic Substances and Disease Registry (ATSDR). 1992. Toxicological profile for Sarin. Atlanta, GA: U.S. Department of Health and Human Services, Public Health Service.

UN Security Council Resolution 687

This resolution was adopted by the UN Security Council at its 2981st meeting, on 3 April 1991. The pertinent section of this resolution, as related to the Khamisiyah report, follows:

6. Notes that as soon as the Secretary-General notifies the Security Council of the completion of the deployment of the United Nations observer unit, the conditions will be established for the Member States cooperating with Kuwait in accordance with resolution 678 (1990) to bring their military presence in Iraq to an end consistent with resolution 686 (1991);

Invites Iraq to reaffirm unconditionally its obligations under the Geneva Protocol for the Prohibition of the Use in War of Asphyxiating, Poisonous or Other Gases, and of Bacteriological Methods of Warfare, signed at Geneva on 17 June 1925, and to ratify the Convention on the Prohibition of the Development, Production and Stockpiling of Bacteriological (Biological) and Toxin Weapons and on Their Destruction, of 10 April 1972;

Decides that Iraq shall unconditionally accept the destruction, removal, or rendering harmless, under international supervision, of:

(a) All chemical and biological weapons and all stocks of agents and all related subsystems and components and all research, development, support and manufacturing facilities;

(b) All ballistic missiles with a range greater than 150 kilometres and related major parts, and repair and production facilities;

Decides, for the implementation of paragraph 8 above [paragraph 6 is only numbered paragraph in document], the following:

(a) Iraq shall submit to the Secretary-General, within fifteen days of the adoption of the present resolution, a declaration of the locations, amounts and types of all items specified in paragraph 8 and agree to urgent, on-site inspection as specified below;

(b) The Secretary-General, in consultation with the appropriate Governments and, where appropriate, with the Director-General of the World Health Organization, within forty-five days of the passage of the present resolution, shall develop, and submit to the Council for approval, a plan calling for the completion of the following acts within forty-five days of such approval:

Reference: UN Security Council Resolution 687, dated April 1991

TAB B - Units Identified as Being Within a 50 Kilometer Radius of Khamisiyah ASP (4-15 March 1991) [97]

The following table shows those units, and reported total personnel strengths, which have been identified by investigators as being present during the demolition operations at Khamisiyah ASP:

MAJOR COMMAND	UNIT DESIGNATION	PERSONNEL STRENGTH
82nd Division (Airborne)	Hqs., 82nd Div	399
	Tactical Command Post (TAC), 1st Bde	122
	Tactical Operations Center (TOC), 3rd Bde	112
	1st Bn, 504th IN	757
	2nd Bn, 504th IN	794
	1st Bn, 505th IN	787
	2nd Bn, 505th IN	778
	3rd Bn, 505th IN	772
	4th Bn, 325th IN	774
	1st Bn, 319th FA	462
	2nd Bn, 319th FA	468
	1st Sqdn, 17th Air Cavalry	772
	3rd Bn, 73rd AR	596
	313th MI Bn	474
	307th Medical Bn	370
	307th EN Bn	498
	37th EN Bn	511
	450 Civil Affairs Bn	77
24th IN Division (Mech)	Main Command Post, 24th IN Div	908
	Hq., 197th IN BDE	323
	2nd Sqdn, 4th Cavalry	404
	24th Signal Bn	668
	724th Combat Support Bn	855
	1st Bn, 5th ADA	635
	Hqs., 36th EN Group	71
	3rd EN Bn	682
	5th EN Bn	807
	299th EN Bn	601
	362nd EN Co	156
101st Airborne Division (Air Assault)	Rear Command Post, 2nd Bde	87
	Hqs., 101st Aviation Bde	146
	1st Bn, 320th FA	436
Other Units	2nd Sqdn, 3rd ACR	866
	Hqs., 265th EN Group	75
	Hqs., 937th EN Group	79
	12th EN Bn	747
	46th EN Bn	605
	264th EN Co	98
	Tactical Command Post (TAC), XVIIIth Corps Artillery (Airborne)	219
	1st Bn 181st FA	443
	1st Bn 623rd FA	411

	Hqs., 513th MI Bde	197
	Hqs., 12th Aviation Bde	146
	9th Chemical Co.	41
	36th Medical Detachment	58
	5th Mobile Army Surgical Hospital	150
	41st Medical Hospital	247
	47th Combat Support Hospital	234
	47th Field Hospital	284
Total		20,867

The following units have been identified to the IAD through contacts with commanding officers. The IAD is providing this information to a separate team whose focus is to verify unit locations :

24th Infantry Division (Mechanized)

1st Bde:
HHC 1st Bde
2/7th IN Bn
3/7th IN Bn
2/69th AR Bn
1/41st FA Bn
5th EN Bn
24th Fwd Spt Bn

2nd Bde:
HHC 2nd Bde
3/15 IN Bn
1/64th AR Bn
4/64th AR Bn
3/41 FA Bn
3rd EN Bn
224th Fwd Spt Bn

197th IN Bde:
HHC 197th
1/18th IN Bn
2/18th IN Bn
2/69th AR Bn
4/41st FA Bn
299th EN Bn
324th Fwd Spt Bn

Division Support Command:

HHC & MMC, DISCOM
724th Support Bn (Main)
91st Chemical Co.
327th Chemical Co. (DECON)
197th Support Bn
82nd Ordnance Det.
83rd Ordnance Det.

Medical:
5th MASH
2nd MASH
10th MASH
274th Field Surgical Team
595th Medical Co.
3/565th Medical Co. (AMB)
47th Cbt Spt. Hosp.
498th Air Ambulance Co.
34th Medical Bn
786 Medical Det. (KA)
702nd Medical Co. (CLR)
690th Medical Co. (AMB)

171st Corps Support Group
260th QM Bn:
110 Supply Co. (POL)
84th Med. Truck Co. (Cargo)
416th Med. Truck Co. (POL)
542nd Maint. Co.
24th Ordnance Co.
851st S&S Co.
548th S&S Bn:
57th Med. Truck Co.
1083rd Heavy Truck Co.
514th Maint. Co.
460th S&S Co.
541st Maintenance Bn
226th Maint. Co.
632nd Maint. Co.
991st Heavy Truck Co.
133rd Ordnance Det.
118th Ordnance Det.

24th Aviation Bde:
HHC 24th Avn Bde
1/24th Avn Bn
3/24th Avn Bn
1/58th Avn Bn (ATC)

Division Artillery:
HHC, DIVARTY
G-333 FA (TAB)

212th FA Bde:
2/17th FA Bn
2/18th FA Bn
3/27th FA Bn
C-25th FA (TAB)

Division Troops:

2/4th Cavalry Sqdn
124th Military Intelligence Bn
36th EN Group
362nd CSE Co.
264th MGB Co.
1/5th ADA Bn

24th Military Police Co.
211th Military Police Co.
519th Personnel Service Co.
24th Finance Support Unit
422nd Civil Affairs Co.
Det. 300 Postal Co.

24th Signal Bn HHC Division

1st Corps Support Command (COSCOM)

46th Corps Support Group (CSG); assigned to the 82nd Div (ABN)
171st CSG; assigned to the 24th ID (MECH)
101st CSG; assigned to the 101st Div (AASLT)

If you are aware of units or individuals who were within the 50-kilometer radius of Khamisiyah who are not listed above, please contact the DoD Persian Gulf Task Force Hot Line at 1-800-472-6719.